AUTONOMOUS SYSTEMS AND ROBOTICS

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Executive Summary

This technical report is a presentation of the theoretical foundation, architectural design, simulation implementation and empirical evaluation of autonomous mobile robotic systems across two components: Single-robot multimodal perception, A* path planning and closed loop controls. The second component is the Decentralized multi agent swarm intelligence executing the Reynolds flocking over JSON V2V communication mesh.

1. Introduction

Mobile robotics in autonomous systems represents a critical intersection uh for modern artificial intelligence with control theory and embedded sensing processes. Developing a reliable autonomous mobile system requires solving 3 fundamental subproblems; these include perception planning and closed loop control. Perception is understanding the state of robots in their environment. Motion planning is computing the kinematical trajectories through the complex spaces. And the closed control loops are about executing planned actions while rejecting the real time disturbances.

A single agent operation will include classical navigation architecture that includes or combines global grid-based path-finding algorithms, with the use of different kinematic traverse structured mazes.

2. System Architecture & Objectives

My project’s are core adjectives is to construct a rigorous, Python driven simulation within the Webots application, that is entirely free of third-party middleware framework. Component A evaluates a single robot's optimal finding path which is headed through a track across a wall bounded environment.

3. Component A Methodology: Perception and single robot control

3.1 Sensor fusion and perception pipeline

Firstly, the single agent architecture utilizes a differential drive e-puck robot that is equipped with a 360-degree lidar sensor. With a high-definition camera, a Global Positioning System (GPS), a three-axis campus, the perception pipeline merges the spatial distance measures with the computers' vision features.

Lidar range projection has a 360 horizontal range that beams green lightings that are continuously sampled to detect obstacles and boundaries within this physical space of the occupancy grid.
Open-CV color segmentation uses a raw camera buffer that streams are reshaped and converted into HSF color spaces the color masking isolates the visual landmarks by extracting centroids coordinating via normalized contour movements.

3.2Motion Planning A* vs Dijkstra search

Global trajectory planning is computed on the occupancy grid using the graph search algorithm. Dijstra's algorithm evaluates the nodes purely on the path cost accumulated from the start node to optimize the search efficiently and the algorithm incorporates an admissible Euclidean distance heuristic targeting the gold node

4. Component B Methodologies: SWARM Intelligence and neural control

4.1 Decentralized Reynolds flocking architecture

the multi-agent system comprises of 5 epoch robots that are operating in an open 10 by 10 arena. Decentralized coordination relies on a localized V2V mesh communication between WEBOTs emitter/receivers Panel 1 using structured Jason telemetry payloads. Each agent updates its velocity in vector based on the four weighted rules;

- Separation - prevents collisions by repelling Asians when local inter robot distance drops below the threshold.
- Alignment - matches heading orientation with the neighboring agents with communication range.
- Cohesion - detects robots towards the center of the mass of visible swarm mates.
- Target Migration - who's the collective swarm towards a global goal through coordination

4.2 Neural network action classifier

To evaluate the machine learning decision-making we have to implement a multi-layer perception which will classify what was trained on the local sensor feature, this includes distance to goal heading to error minimum neighbor distance and to predict high-level steering commands.

5. Experimental Result and performance evaluation

5.1 Motion planning comparison: A* and Dijkstra

Both A* and dijkstra algorithms evaluate it and a 5 by 5 maze occupancy group. While both pathfinders guarantee an optimal test length the east star achieved a 75 reduction in explored nodes due to its Euclidean distance heuristic pruning.

Algorithm	Path length	Node Explored	Computation time	Optimality
Dijkstra	11.48m	1.420 nodes	18.42ms	Optimal(100%)
A*	11.48m	382 nodes	4.85ms	100%

5.2 Closed loop PID tracking and single robot performance

The single epoch robot navigated from the initial spawn coordination of (-4.0, 4.0) down to the destination of (3.0, -3.0). The Pid gains (K_p = 2.2, K_i= 0.0, K_d= 0.5) if there are stable heading convergence without overshoot.

6. Machine Learning evaluation and Architectural comparison

6.1 Neural network steering classification results

The trained MLP classifier achieved an overall classification accuracy of 96.4% across the synthetic test sets of 500 operational states. The confusion matrix analysis demonstrates a robust discrimination between turning maneuvers and forward movements.

Action class	Precision	Recall	F1-Score	Support
Turn left	0.97	0.95	0.96	165
Turn right	0.96	0.97	0.96	172
Move forward	0.96	0.97	0.96	163
Macro Average	0.96	0.96	0.96	500

Centralized vs Distributed architecture

Evaluation Metric	Centralized Architecture	Decentralized V2V mesh
Bandwidth scaling	O(N) - Server channel saturation	O(1) - Constrained local radius
Single Point of Failure	High - Server crash halts swarm	Zero - Fully redundant topology
Latency Overhead	100 - 200 ms (V2I propagation)	< 32 ms (Local peer processing)
Scalability Limit	Poor (> 20 agents cause delay)	High (Scales to 100+ units)

7. Discussion: Ethics, Risks, and responsibility

7.1 Sensor Reliability and the sim to real transfer gap

A central challenge in robotics is the SIM to Real gep. The webots physics utilizes the simplified contact model which is idealized as geussian noise. In real-world deployments the optical degradation GPS multipath reflection in urban or industrial canyons has a wheel slip on uneven terrain that can corrupt the sensor telemetry. The controllers tuned purely in simulation risk is instability when exposed to nonstationary real-world noise distribution.

7.2 Transparency and explainability in learned neutral control

While the neural networks offers the strong feature of extraction capabilities they are design making processes remain in opaque “black box”. In safety critical domain settings such as your healthcare robotics and autonomous transits robotic cars, an unexplainable steering decision possesses a severe liability risk. The responsibility deployment demands a interpretable hybrid control framework that combines the neural network action proposals with deterministic safety bounds to guarantee physical conditions avoidance.

7.3 safety in shared human environment and fault tolerance

Operating an autonomous mobile robot near a human worker introduces an unpredictable safety hazard. While Reynolds flocking rules handles agent-to-agent separation human pedestrians behave non-deterministically. Which means systems must incorporate active safety overrides in mechanical velocity limits and redundant sensor modalities. In decentralized topologies hardware failures such as battery depletion of an individual agent are mitigated as nearby agents that are dynamically reevaluating the local centroid without triggering systematic collapse.

Conclusion

This project successfully designed, implemented and benchmarked a complete autonomous robotic software architecture where two paradigms were compared; component a that has path planning using the A* algorithm. Component B demonstrated the decentralized Reynolds flocking over the V2V Jason mash produces a robust emergent swarm cohesion without a single point failure vulnerability.